I

Beyond Visual Range Chronicles

Minimum Abort Range

v. 3

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Abstract

This study was done using Wikipedia data or TacView data taken from Falcon 4 BMS 4.33 U5 encounters. It tries to set up an easy to understand picture on the Minimum Abort Range computation and concept. As this is neither an actual tutorial on BVR maneuvers nor a fluid Dynamics lesson[1], unaccounted for scenarios can induce differences from what you, the reader, encounter. That being said, the methods and plots should give you a nice explanation of what the most successful missile avoiding tactic is all about. All cases are considered to take place around 25'000 ft altitude but that cannot be hard set because of how dynamic this scenario is. Furthermore, the missile algorithm can make it shoot up after launch and do most of it's "cruise" at high altitudes (mostly the AMRAAM and the Phoenix), but that's what makes this analysis even better, it also studies "actual" encounters and not just theoretical ones.

In v. 2, I've added the AIM-54 Phoenix long range active radar homing missile by doing some F-14B encounters.

In v. 3, I've revisited the AIM-54 Phoenix case III using the Heatblur's F-14B for DCS. All three variants of the Phoenix were tested.

Contents

1 Definition **3**

2 Missile flight **3**

2.1 Missile Aerodynamics 3

2.2 Bandit's launch speed 4

2.3 My speed 4

2.4 Altitude 4

2.5 AA-x vs. AA-x 4

[1] I tried to keep it in the kinematics sphere as much as possible

3 *MAR* determination **5**
3.1 Case I
MAR based on maximum operational range (MOR) - uniform deceleration . . . 5
3.1.1 Missile breakdown . . . 7
3.1.2 Comparison . . . 8
3.2 Case II
MAR based on maximum operational range (MOR) - variable deceleration . . . 10
3.2.1 Missile Breakdown . . . 12
3.2.2 Comparison . . . 12
3.3 Case III
MAR based on TacView encounters . . . 15
3.3.1 Missile Breakdown . . . 15
3.3.2 Comparison . . . 18

4 Conclusion **20**

1 Definition

The Minimum Abort Range (*MAR*) is the closest distance to the bandit that you can get and still be able to turn and run from his/her missile. This distance ensures that you have a large enough lead on the missile that, by the time it reaches you, it stalls out because it lost too much speed. This is called beating the missile **kinetically** as it cannot follow you anymore because it only falls out of the sky.

2 Missile flight

A BVR missile flight starts off at the launch, at which point its' motor is fired and the missile starts accelerating until the motor burns out (6-15 seconds). After that, the missile starts decelerating because of drag. Missile will decelerate less when it just flies straight and more as the G load increase in turns. Nevertheless, it always decelerates. The magnitude of the deceleration is the most important factors behind the *MAR*.

Even more interesting is the straight and level deceleration (1-G deceleration), as in a pure turn-and-run tactic you're not really putting too much maneuvering on the missile. The factors affecting *MAR*are:

1. missile aerodynamics
2. bandit's launch speed
3. my speed
4. altitude

2.1 Missile Aerodynamics

This factor is strongly related to the actual deceleration of the missile in un-powered flight. Some missiles lose less speed than others in the same amount of time so the *MAR*s differ. Actually, even when using the same missile with different other factors can yield totally different results. This factor also influences how much deceleration is added during turns as some missile can only do coordinated turns while others can stall-turn (slip turn) and then reacquire coordinated flight. Most modern IR, short-range, missiles stall-turn because the engine burn usually lasts throughout the entire flight (5-7 seconds to hit the target).

A stall turn is not normally in a BVR missile's "inventory" but there are exceptions like the R-77[2] with grid fins that can turn $150^o/s$. How do I know this is a slip turn? Because if this turn would be coordinated at Mach 2 it would mean the missile is pulling 161 Gs which it cannot do. And this missile can do Mach 4! But this feature of the R-77 was maybe added because there is an IR seeker variant (R-77T) which needs the maneuvering in the later stages of flight. However, the problem with a stall-turn is the high drag it induces because of the huge AoA.

In any case, missile aerodynamics influences its' drag and acceleration/deceleration.

[2]AA-12 Adder

2.2 Bandit's launch speed

The speed of the launching platform is highly linked to the speed of the missile when the engine burns out. And because the missile decelerates continuously, the burn-out speed increases the *MAR* as there is more speed to bleed of before it stalls. It is one thing to launch an AIM-120 from a Mach 1.5 F-15 flying at 40'000 ft and a whole other thing to launch the same missile from an F-18 doing Mach .7 at sea level.

Suffice to say that the higher the initial launch speed, the higher the starting speed in unpowered flight and the higher the *MAR*.

2.3 My speed

My own speed influences the *MAR* as increasing my own speed makes the missile approach me slower. For example, if I abort a missile running away at 700 kts the missile will start off catching me at 1'100 kts and decelerate all the way. If I only run away at 400 kts the missile starts catching up with at 1'400 kts. This 300 kts difference may mean you are dead.

The faster you go, the faster you run away from the missile so the smaller the *MAR* gets.

2.4 Altitude

Altitude is a factor that influences drag, so deceleration. In a somewhat simplistic view, drag on a flying body equals:

$$\frac{C_L A_R v^2 \rho}{2}$$

, where ρ is the density of the surrounding air which is dependent on the altitude of the flight. So drag is inversely proportional with altitude[3]. From a turn-and-run point of view, it is thus better to decrease altitude for the missile as that means higher drag and smaller *MAR*.

2.5 AA-x vs. AA-x

As you can see, there are quite a few factors influencing the *MAR*. For the same missile - same factor 1 - the other three factors can get into such a setup that in one instance the *MAR* is 20 nm while in another it can barely reach 7 nm.

On the other hand, two different missiles (AIM-120C and AIM-7M) can have the same *MAR* by getting the factors right.

For the Attacker

As an attacker, you would want to increase the *MAR*. As the 1st factor is fixed, to increase the *MAR* you have to:

[3] density of air decreases as altitude increases

- increase speed before launch
- increase altitude

For the Defender

As a defender, you would want to decrease the *MAR*. As the 1st factor is fixed, to decrease the *MAR* you have to:

- increase speed
- decrease altitude

3 *MAR* determination

The *MAR* is highly related to the deceleration of the missile. The hardest thing to estimate in these theoretical determinations is the final velocity of the missile because it is related to the missile's ability to still fly and maintain altitude. What is the stall speed of a missile? And bear in mind that at BVR altitudes (>20'000 ft) there is a significant difference between IAS (what keeps wings flying) and TAS (what makes a missile approach a target). So if a missile stalls at Mach .6 , it means that at sea level it stalls out at $\sim$ 400 kIAS and kTAS[4]. At 25'000 ft however, the stall speed of 400 kIAS equals $\sim$ 600 kTAS.

Of course, this is related to actual missile aerodynamics which I'll not get into any detail. Suffice to say that large fin missiles stall at lower speeds than small fin ones, the caveat being that they also decelerate faster because of higher drag. In our determinations, however, we'll use the same stall speed for a missile of 400 kIAS which relates to 600 kTAS at 25'000 ft, where we'll consider all our fights take place[5]. But what's even more interesting is that we don't even care about the actual stall speed of the missile. A 600 kTAS missile can be considered beaten even if it can still fly at 300 kTAS. Furthermore, the target speed is important as in the turn-and-run tactic, if the missile moves slower than the target, it can be considered stalled. So we'll use 600 kTAS as the stall speed for encounters with targets slower than 600 kTAS, and the target speed as the stall speed if it is greater than 600 kTAS.

3.1 Case I
MAR based on maximum operational range (MOR) - uniform deceleration

The most useless parameter in BVR is the maximum operational range of a missile reported by the manufacturer or various info Internet sites. Needles to say that a *MAR* based on

[4]missiles stall out at higher speeds than normal winged flying objects because they use small fins as wings; also, at these low speeds the missile is basically beaten as it cannot catch up with any fighter

[5]we'll also consider an International Standard atmosphere

this parameter is pretty useless, but it has a huge teaching value as a first dive into the *MAR* determination.

The scenario for computing the deceleration in the maximum operational range setup with uniform deceleration is that at the moment of launch, the missile is instantaneous at maximum speed and the target is cold on the missile. The motion of the missile is considered uniformly accelerated (decelerated) and rectilinear. So the method consists in using the equations of motion to get the constant missile deceleration value. We then compute the distance the missile is traveling before reaching the stall speed and the time of flight and subtract from that the distance the target traveled in that same time. Don't forget that the stall speed of the missile is 600 kTAS for targets speeds lower than 600 kTAS, and equal to the target speed otherwise.

The caveat of this approach is that throughout the whole missile flight, the acceleration is considered constant. This is opposite to what happens in real life as the faster the missile goes, the higher the deceleration. In case II we'll consider a more true to life acceleration in which the deceleration is derived from the speed of the missile.

Another simplification happened when we considered the missile speed as maximum operational speed immediately after launch. In real life, the missile motor has a burn time in which the missile is fastly accelerating to the maximum speed (or close to it) in the region of 6-10 seconds. At case III we'll actually take into account the acceleration segment of the missile, too.

The constant deceleration value is computed using the equations of motion and by solving the following system of equations for a and t.

$$d = d_0 + v_0\, t + 1/2at^2$$

$$v = v_0 + a\, t$$

where d = MOR, d_0 = 0, v_0 = maximum missile speed and v = missile stall speed. This results in $a = \frac{v^2 - v_0^2}{2\, d_0}$ while $t = \frac{2\, d}{v_0 + v}$ or in a more intuitive form:

$$a = \frac{v_{STALL}^2 - v_{MAX}^2}{2\, MOR}$$

$$t = \frac{2\, MOR}{v_{MAX} + v_{STALL}}$$

Having the deceleration of the missile, I can compute the distance that the missile travels to reach the actual stall speed used[6] by using the space equation of motion again:

$$d_{missile} = v_{MAX}\, t + 1/2\, a\, t^2$$

where a was computed above and $t = \frac{v_{STALL} - v_{MAX}}{a}$[7]. Needless to say that for target speeds lower than 600 kTAS, this distance equals the MOR.

The distance the target is traveling is pretty straight forward as it has no acceleration:

$$d_{target} = v_{target}\, t$$

where t is the same $t = \frac{v_{STALL} - v_{MAX}}{a}$ missile flight time. The *MAR* is thus given by:

[6] 600 kTAS or target speed

[7] from $v = v_0 + a\, t$

$$MAR = d_{missile} - d_{target}.$$

I applied this method for a few very used missiles in the virtual skies.

3.1.1 Missile breakdown

The missiles used for this analysis are: the AIM-120C AMRAAM, the AIM-120B AMRAAM, the AIM-7M Sparrow, the R-27ER/AA-10C Alamo, the R-77/AA-12 Adder, the Matra Super 530D and the AIM-54 Phoenix long range missile. The parameters used for the missiles are presented in table 1. I am not going to reflect much on the Phoenix for this case as it has the most MOR and the least deceleration so, by these standards alone, it definitely is the best missile ever. Things tend to decay in case II and III quite a lot.

	v_{MAX} **[kTAS]**	MOR **[nm]**	a [$^{kts}/_{s}$]
AIM-120C	2'408	60	-13
AIM-120B	2'408	40	-19
AIM-7M	2'408	38	-20
R-27ER	2'709	70	-14
R-77	2'408	45	-17
530D	2'709	20	-48
AIM-54	3010	100	-12

Table 1: Missile flight parameters for the uniformly decelerated case

A word on acceleration unit of measure which may seem off for some readers. I use knots per second as opposed to meters per squared seconds or other more usual unit because I see it as the most intuitive one. An acceleration of $-13\ ^{kts}/_{s}$ means that every second the speed of the missile drops by 13 knots. This is much more intuitive than replacing it with $\sim -7\ ^{m}/_{s^2}$.

	MAR **[nm]**						
target TAS [$kts/Mach\#$]	**AIM-120C**	**AIM-120B**	**AIM-7M**	**R-27ER**	**R-77**	**Super 530D**	**AIM-54**
500 / .83	39	27	25	48	29	14	73
602 / 1	35	24	23	44	27	13	67
722 / 1.2	30	21	20	39	23	11	61
903 / 1.5	23	16	15	31	18	9	51
1'204 / 2	12	8	8	19	9	7	38

Table 2: Different *MAR* values for different missiles and target speeds in the uniform deceleration scenario

So what's the 530 doing here? Well, the Mirage-2000C is quite a big deal[8] in DCS World so it would have been a big miss if the 530D wasn't here. While it aligns along the lines of the

[8] for me

Sparrow in terms of guidance, with a larger initial speed but a lower range, it definitely has a thing or two to say in the BVR arena.

3.1.2 Comparison

Figure 1 plots the *MAR* for different target runaway speeds based on the maximum operational range of the missile. From this plot, we can see that the most standoffish missile is the R-27 due to its' higher range and maximum speed. The AIM-120C comes second while the 530D is the worst in keeping a bandit at bay, expected because of its' low MOR. The AIM-7M is the second "worst" missile but it closely matches the R-77.

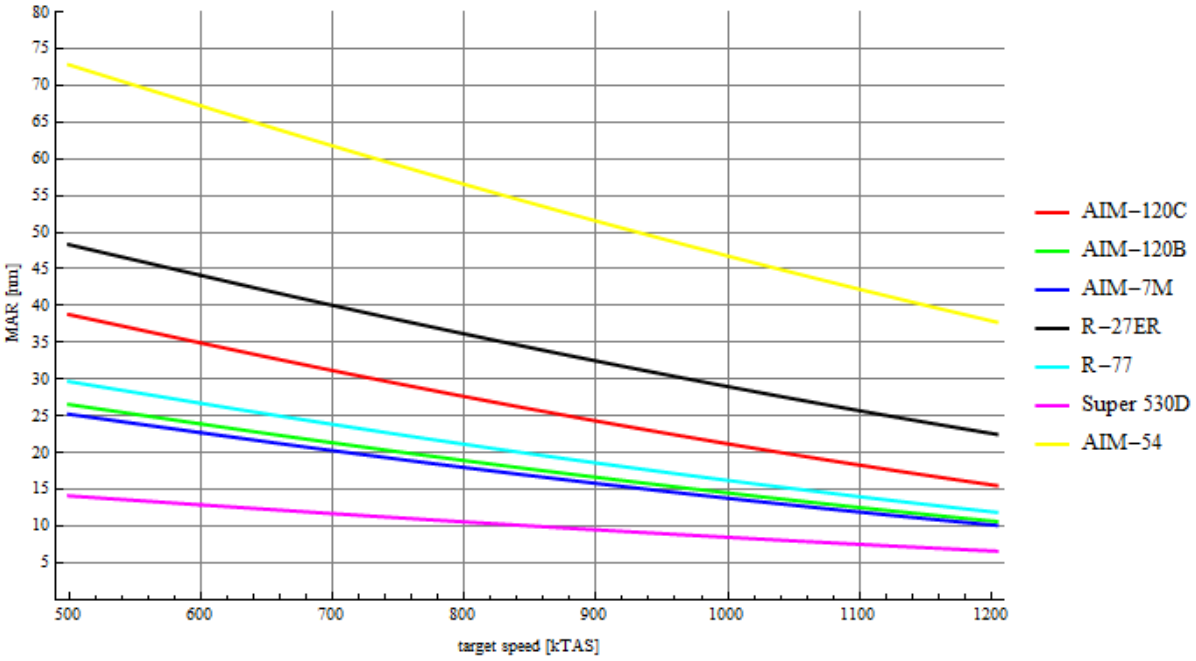


Figure 1: *MAR* from MOR comparison

Another interesting comparison that I can make is the percentage of maximum *MAR* that the missile provides as a function of target speed. The maximum *MAR* is defined as the *MAR* computed with the above method for a target running away at 500 kTAS. This kind of comparison would show how fast the missile is loosing the battle with the target in a turn-and-run scenario. Figure 2 shows this comparison.

The 530D is not looking as bad now, is it? At 1150 kTAS target speed, this missile still retains 50% of its' maximum *MAR*. Meanwhile, the R-77 can only reach about 43% of its' maximum *MAR*. Actually, behind that cyan line, besides the R-77 there are the AIM-120C, AIM-120B, AIM-7M. So even the AMRAAM-C, is inferior to the 530D and the R-27ER in keeping the lethal radius high as target speed increases. This is due to the higher initial speed of these two missiles. At usual target speeds (700 kTAS), all missiles are matched in this parameter.

Yet another comparison, maybe more interesting even, is the percentage of *MAR*to MOR that the missile retains as target speed increases. Figure 3 shows this comparison. The most

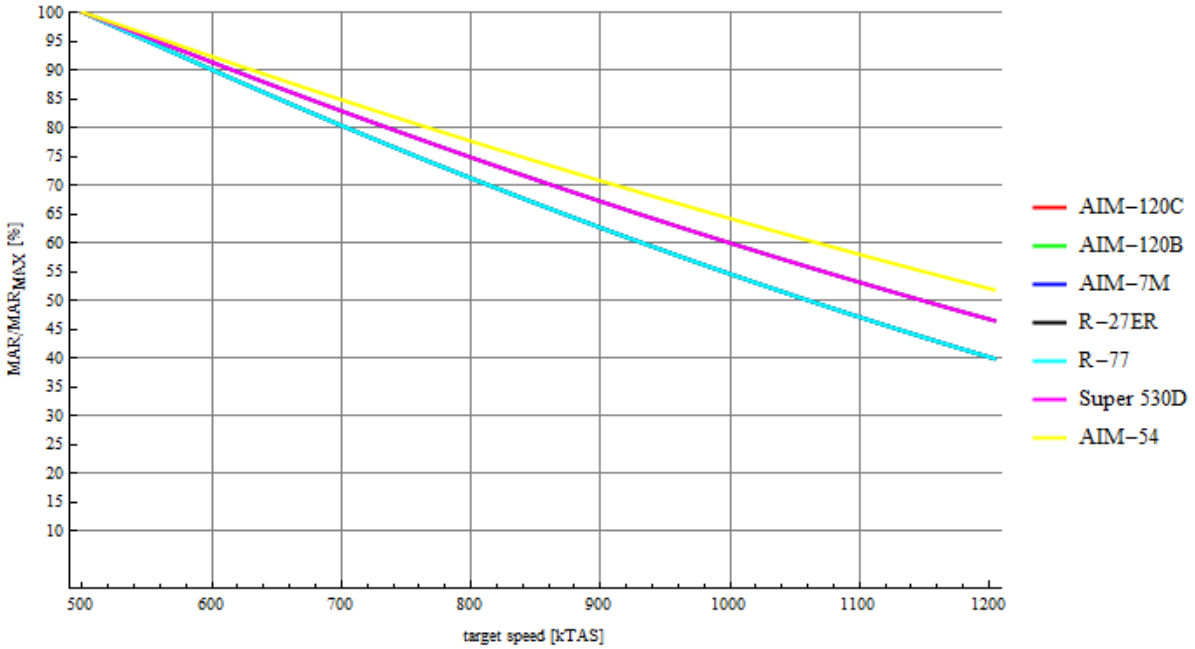


Figure 2: *MAR* from MOR normalized to the maximum *MAR* comparison

lethal missile throughout its' MOR is the 530D as for a slow target it reaches more than 70% of its' MOR. Yes, it has a very low MOR but is good to know that you have a very high probability of hit within that entire MOR than you have with the rest of the missiles.

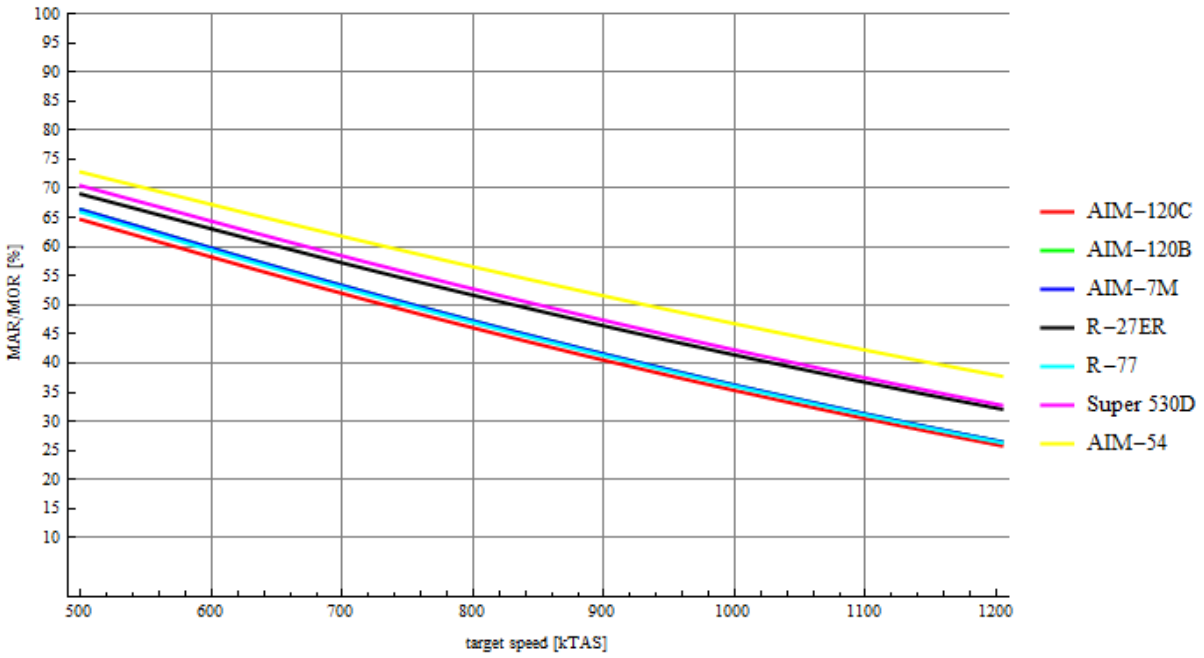


Figure 3: *MAR* from MOR normalized to the MOR comparison

3.2 Case II
MAR based on maximum operational range (MOR) - variable deceleration

This is the same *MAR* determination based on MOR, but this time the acceleration si more true to life as it variates during the whole flight. The parameter that stays the same as in the uniformly accelerated version is the MOR - the missile should still reach the MOR by the time it gets to v_{STALL}.

I divided the missile flight in smaller segments of constant speed where the actual segment speed is dependent on the previous speed and the deceleration on the current speed of the missile. But how does a missile decelerate based on speed? As much as this is a question on fluid dynamics, I can still give some educated, purely kinematic answers. We know that in a simplistic form drag is $\frac{1}{2}C_l A \rho v^2$, which means that drag is directly proportional with the squared speed. So I am computing the deceleration based on the squared speed with the condition that at the highest speed the drag is a value measured in TacView, while at stall speed deceleration is 0. So the natural equation that comes to mind is a variation of the rule of three:

$$a = a_{max} \frac{(v_{missile} - v_{stall})^2}{(v_{max} - v_{stall})^2}$$

where, in this case, v_{stall} always equals 600 kTAS as the deceleration is not dependent on the speed of the target. This gives a closer to life deceleration of the missile. To get a feel of how the acceleration profile looks like in this case, I've plotted the profile in figure 4. Low deceleration is at the top and increases towards the bottom because the acceleration values are negative.

The "granularity" of the method I used is given by a 1-second window segment, which means that the missile has a constant speed for every second of flight. It's not a continuous dependency on speed, but good enough.

A small note on maximum deceleration via actual TacView encounters. While I am trying to use scenarios where the missile reached its' maximum published speed, in actual encounters this may not happen for different reasons[9]. Also, because of how I read the speeds of the missile, maximum deceleration may not be reached in the first second of flight but the second or the third. Because of this, initial maximum deceleration value is the average value of the first three seconds of flight after maximum velocity. But if a missile didn't reach its' maximum operational speed, I do a simple extrapolation of the 3-value averaged deceleration to the maximum speed. For example, my chosen AIM-7M scenario has a maximum attained speed for the missile of $1'902\ kts$ from a $2'408\ kts$ possible. The maximum deceleration is $-16\ {}^{kts}/_{s}$. Using the three simple rule with a square dependency we get the extrapolated maximum deceleration:

$$a_{max} = \frac{v_{max}^2}{v_{measured}^2}\ a_{measured}$$

which results in $-27\ {}^{kts}/_{s}$ maximum deceleration for the Sparrow. The method is coherent from the point of view that at the actual measured maximum speed, the deceleration will be spot on with the measured value.

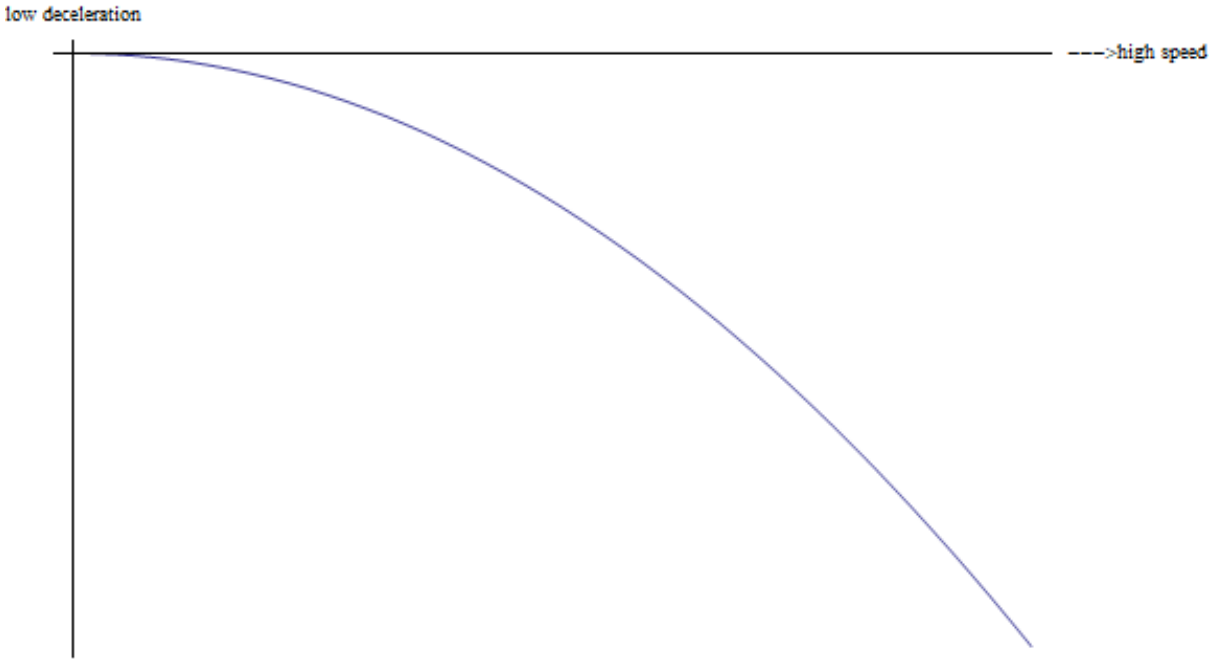


Figure 4: Deceleration profile in a squared speed dependency

The method is then simple enough: I compute the distance traveled by the missile with the computed variable decelerated speeds and subtract from it the target distance traveled in the same amount of time. For this I've made a small program that computes everything based on maximum operational speed, initial deceleration, MOR of the missile and speed of the target.

[9]e.g. not in ISA conditions, slow launch speed, hard initial missile turn, etc.

3.2.1 Missile Breakdown

The kinematic parameters used for this case are depicted in table 3.

	v_{MAX} **[kTAS]**	MOR **[nm]**	a_{MAX} [$^{kts}/_{s}$]
AIM-120C	2'408	60	-49
AIM-120B	2'408	40	-68
AIM-7M	2'408	38	-27
R-27ER	2'709	70	-77
R-77	2'408	45	-77
530D	2'709	20	-95
AIM-54	3010	100	-55

Table 3: Missile flight parameters for the variable decelerated case

In table 4 I've put some *MAR*s computed in the Case II scenario. Some missiles got better at catching targets while other got a little worse. The AIM-54 got quite a *MAR* hit because of the much higher deceleration. Nevertheless, it ca still reach its' MOR so don't feel to sorry for it. Case III will show an "actual" engagement.

	MAR **[***nm***]**						
target TAS [$kts/Mach\#$]	**AIM-120C**	**AIM-120B**	**AIM-7M**	**R-27ER**	**R-77**	**Super 530D**	**AIM-54**
500 / .83	36	25	28	41	26	15	64
602 / 1	32	22	26	35	22	14	57
722 / 1.2	26	18	23	28	18	12	48
903 / 1.5	17	13	19	17	11	10	35
1'204 / 2	8	6	13	9	5	7	19

Table 4: Different *MAR* values for different missiles and target speeds in the variable deceleration scenario

3.2.2 Comparison

In figure 5 I'm showing the dependency of the *MAR* on target speed based on the variable acceleration method described above. We can see that the most standoffish missiles are the R-27ER and the AIM-120C. There is however a surprising conclusion for the AIM-7 in that if the target is very fast (> 840 kTAS, Mach 1.4), then the Sparrow is a better choice for firing on a runaway target due to its' lower deceleration. For example, a 1'100-knot target could be hit from almost 15 nm out as opposed to only 10 nm out with an AMRAAM or the Alamo-E. The 530D takes last place as the high starting deceleration and the much lower maximum range take their toll. That being said, at very high target speed (> 1'100 kTAS), the 530 is making a comeback surpassing AMRAAM-B and the Adder.

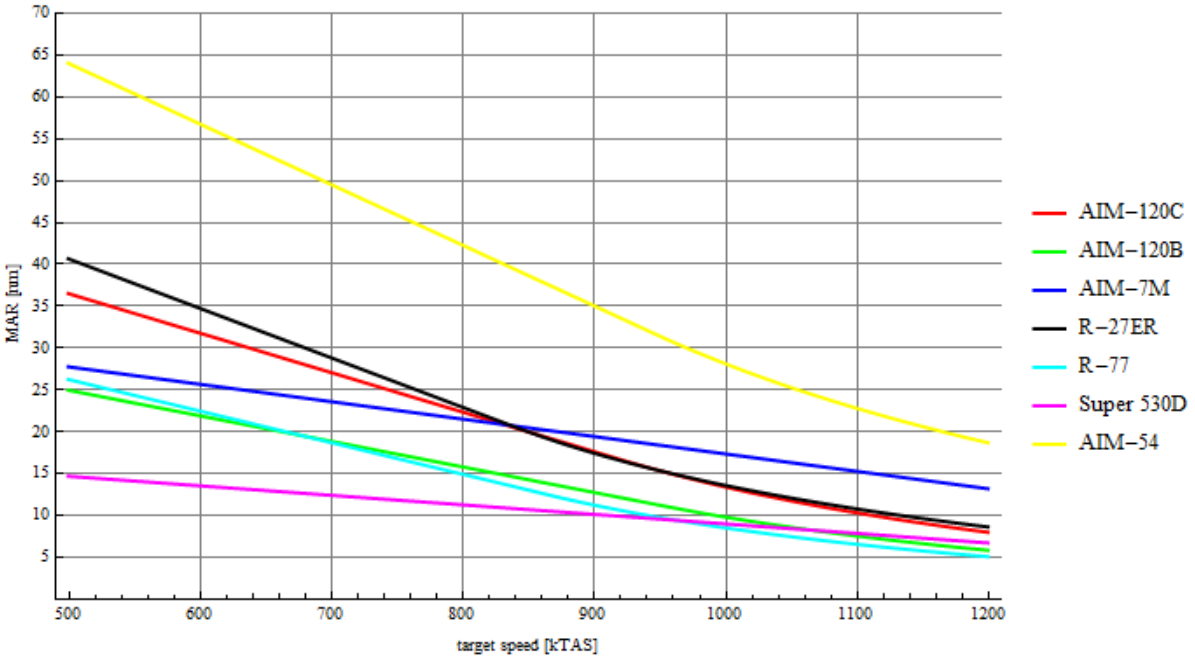


Figure 5: *MAR* from MOR comparison in a variable acceleration setup

Following the comparisons done for the MOR-uniform acceleration case, let's look at the rate of maximum *MAR*plot drawn in figure 6. The best missiles at maintaining their *MAR*s through a wide target velocity range are the AIM-7M Sparrow and the Super 530D. These are also the smallest MOR missiles and are both semi-active radar homing. Even so, their *MAR*s still take more than a 50% hit. But the other missiles *MAR* degrade with about 80%.

From the ratio to MOR plot (figure 7) other interesting conclusions can be drawn. The Sparrow and the 530D, in the range of normal BVR fights[10], retain more than 60% of their MOR as *MAR*. This is impressive as the other more advanced missiles only retain less than 50%. At very high target speeds, the discrepancy between the small range missiles and the advanced missiles grows to 20%, with the latter only reaching 15% of the MOR. One could think that this is normal as the difference in absolute MOR will make up for these percents but 15% of 60 nm (=9 nm, AIM-120C) is less than 35% of 38 nm (=13 nm, AIM-7M).

That is not to say that the Sparrow or the 530D are the best missiles to carry. Far from it! The mode of employment has a very heavy word on how successful a missile is in actual engagement. The fact that the R-27ER can actually be launched from 70 nm away (MOR based scenario) just to put the bandit on the defensive before he/she gets a firing solution is a huge advantage. The 530D, on the other hand, with its' 20 nm MOR means that if you fly the Mirage-2000C you should play a superb waiting game and defensive tactics to be competitive in the BVR scene because chances are you will be fired upon first. But, then again, if you are presenting a 900 kTAS target (Mach 1.5) to an AMRAAM-C, the turn and run tactic will annul the bandit's range advantage and level the field, at least in the first-shoot department.

[10]500-700 kTAS

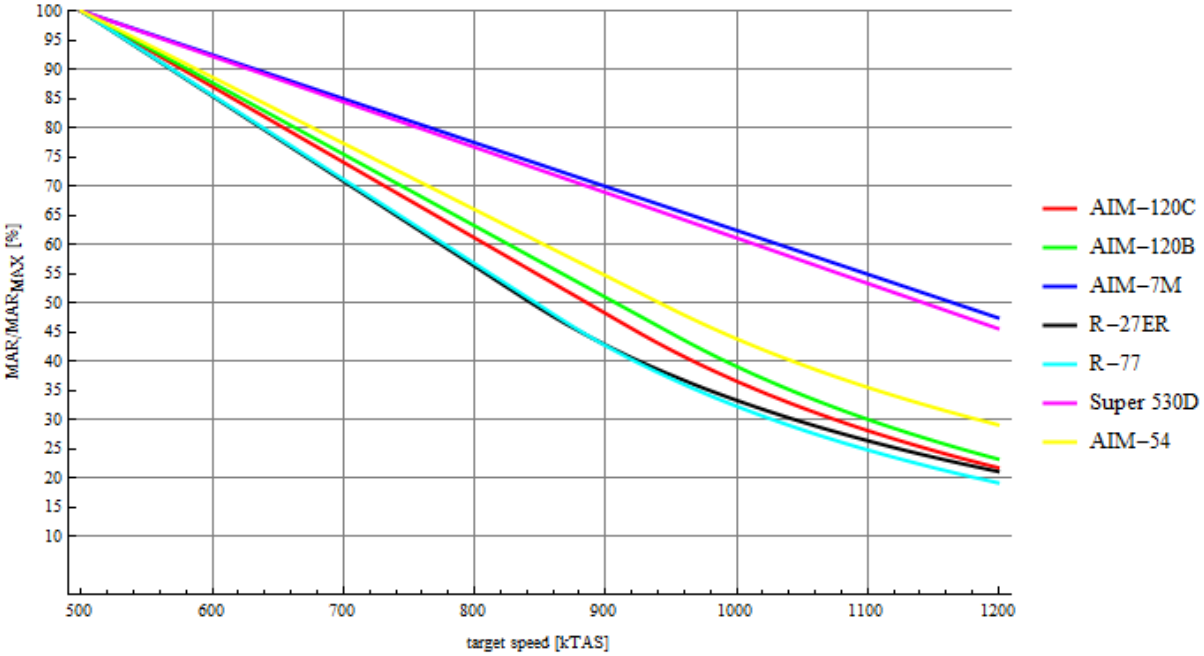


Figure 6: *MAR* from MOR for the variable acceleration scenario, normalized to the maximum *MAR* comparison

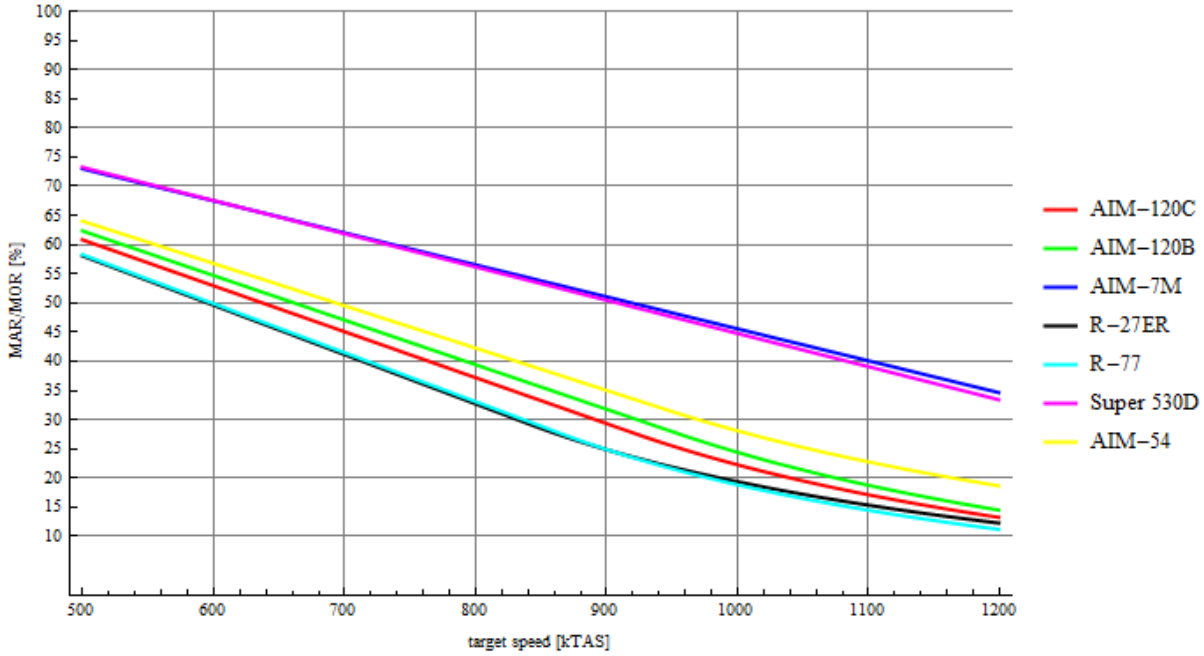


Figure 7: *MAR* from MOR for the variable acceleration scenario, normalized to the MOR comparison

3.3 Case III
MAR based on TacView encounters

This is a much more realistic scenario. During my BVR training I use a lot of TacView analyses to see what went wrong and what went right. So by visually inspecting the encounters I can measure with some degree of precision various parameters, like in the variable acceleration case above. While in the above case I had to measure the deceleration in the first seconds of un-powered flight, in this case I will actually measure the speed of the missile until it stalls out, for each second of flight, in turn and run scenarios.

Having the second by second velocity of the missile gives me acceleration and deceleration analysis and thus more true to life distances for the missiles. The method resembles the other two in the sense that a subtraction between the missile distance and the target distance is done. But this time we actually have the speeds of the missile so we are basically using a more real deceleration distribution, and thus it should result in a more accurate *MAR* computation. But maybe the best thing about this method is the fact that there is no link to the MOR.

3.3.1 Missile Breakdown

As stated above, the "only" missile information I have are the sequential speeds throughout the flight. But, because they are taken from an actual encounter in "real"[11] conditions, it should give more useful results. So I've put the speed evolution for all 6 missiles on the same graph just to get an idea of how these missiles behave in figure 8.

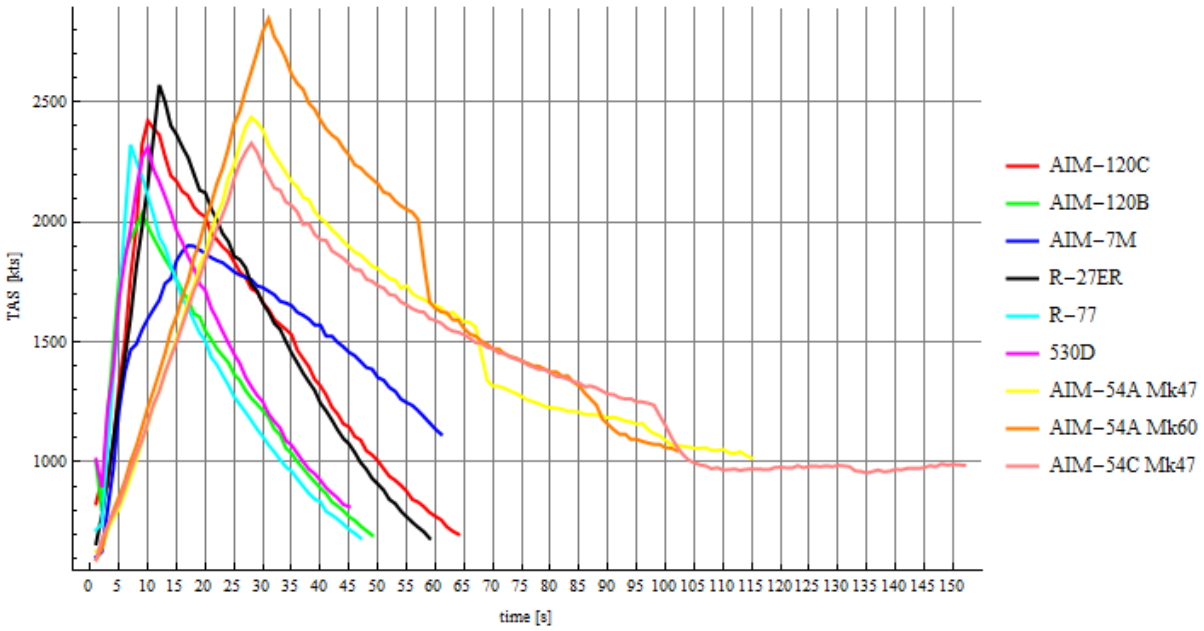


Figure 8: Speed evolution for the six missiles studied in the TacView scenario

[11] virtual

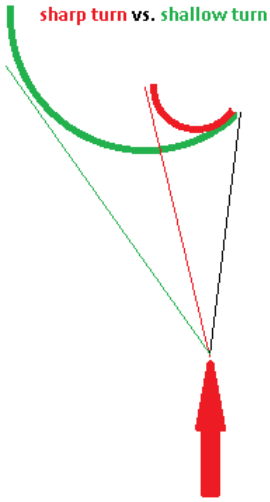


Figure 9: Turn effect on the initial missile trajectory

One interesting fact is that the AIM-7M is actually accelerating in what can be viewed as a two stage rocket motor. And that is true to real life as the AIM-7M has a Hercules MK 58 dual-thrust solid rocket meaning it has two different propellants burning at different rates. Initially, both propellants give it a high thrust. As the fast burning propellant burns out, the missile only relies on the slower burning one so a decrease in acceleration is expected, just as in the plot. This is a nice confirmation that the simulator developers are providing superb flight models. Wow!

Another thing that one can see in the plot is that the missiles didn't go to full speed (2'408, 2'709 and 3'010), usually topping lower or even higher (R-27ER goes to 2'571 kTAS in the plot while the Phoenix doing "just" 2'012 kTAS). That is because I try to simulate the real weather at the time of the encounter in my missions. So I don't get ISA conditions, most of the time. Another reason for this is that in actual encounters you are usually hot on the bandit as he fires and you have to do a turn to abort. That commands the missile to also turn which eats up some of the initial speed. The least effect would be by doing a sharp turn, but in most cases I do a shallow turn (< 4 Gs) to put Gs on the missile, while the AI does sharp turns (> 7 Gs). Figure 9 shows the effect of the two different abort turns on the pursuing missile.

Another grieve is that some missiles have a "high cruise" trajectory. Right after launch, while aided by their rocket motors, some missiles shoot up in altitude to decrease drag and increase range. The exponents of this tactic are the AMRAAM and the Phoenix. While the AMRAAM shoots up 5'000 to 10'000 ft for a 25'000 ft target, the Phoenix went to 42'000 ft for a 30'000 ft target so its' maximum kTAS (Mach 5) could have only get to about 2'800. And this is a big source of discrepancy. But why would I want to divert from a "real" encounter just to get the best that a missile can do when in our virtual skies no sane person lets their bandit have any advantage! This was an actual encounter and I present it as it was![12]

And yet another perceived caveat is the stall speed, on the other side of the spectrum. As

[12]I have to mention that the missile went for a FL300 Sentry AWACS doing its' job and didn't get it because the AI pilot knew better

one can see not all missiles got to 600 knots at the end. That is again because of the "real" scenarios that the readings were measured in. The missiles stalled faster than in the theoretical cases and at different speeds, which is superb as that really gives us a nice separation between the actual missiles, a feature we didn't have for the theoretical cases (all missiles stalling at the same speed).

But analyzing figure 8 gives us other true-to-life parameters which makes me think I'm, at least, on the right track. For example the motor burn times of different missiles. We can see that the R-27ER burns for about 12 seconds which is spot on! The AIM-120C and -B for 10 seconds but we can see a drop in acceleration for that last second so the actual burn time was more closer 7 to 9 seconds. This is consistent with actual real life data of the AMRAAM.

Another thing to notice is that the fastest accelerating missile is the AMRAAMsky, the R-77, which also has a high deceleration. While it is an active radar homing missile, it is made for highly maneuvering targets and escape within the *MAR* is mostly futile as it can achieve $150^o/s$ turns. Nevertheless, outside the *MAR*, it si a wasted launch.

The most speed bleeding missile is the R-27 (most downward slope of the curve). While it could get to huge ranges, the large fins on it make it drop speed very fast even in the shallowest of turns.

The least decelerating missile of them all, and the one which burns for most of the time ($\sim 16s$) is the Sparrow-M. As it entered service in 1982, it is definitely an interceptor's missile. It bleeds very little speed to catch those high and fast flying tactical bombers.

So let's see some values of the *MAR*s in different target speed scenarios computed in the "real" encounter scenarios.

	MAR **[*nm*]**						
target TAS [*kts*/*Mach*#]	**AIM-120C**	**AIM-120B**	**AIM-7M**	**R-27ER**	**R-77**	**Super 530D**	**AIM-54**
500 / .83	17	11	17	16	10	12	32/34/36
602 / 1	15	10	15	14	9	11	28/31/32
722 / 1.2	13	8	13	12	8	9	25/28/27
903 / 1.5	10	6	10	10	6	7	19/23/19
1'204 / 2	6	3	5	6	3	4	9/14/6

Table 5: Different *MAR* values for different missiles and target speeds in the TacView encounters scenario

Table 5 shows the computed values for the *MAR* in the TacView encounters scenario.

The AIM-54 is by far a true BVR stand-off weapon reaching as much as 36 nm *MAR*s. This was expected because that is the true scope of this missile.

The thing that strikes me is how the field for the other missiles is somewhat leveled in this scenario. Yes, there are the AIM-120C, the AIM-7M and the R-27ER that can almost reach 18 nm for slow targets and even for Mach 1 targets they are still true BVR missiles while the rest can barely reach the 12 nm mark for slow targets and WVR for the rest of the spectrum. But there are no huge differences in *MAR*s and remember that every pilot has his/her own abort response to a missile launch warning which makes these values variate. For example if a 700-kTAS pilot responds with a turn after 5 seconds from the warning, it increases the *MAR*s by

about 1 nm. As stated, a shallow turn will also decrease the missile's *MAR* while a turn into denser air (downward in the vertical) would have the same effect.

3.3.2 Comparison

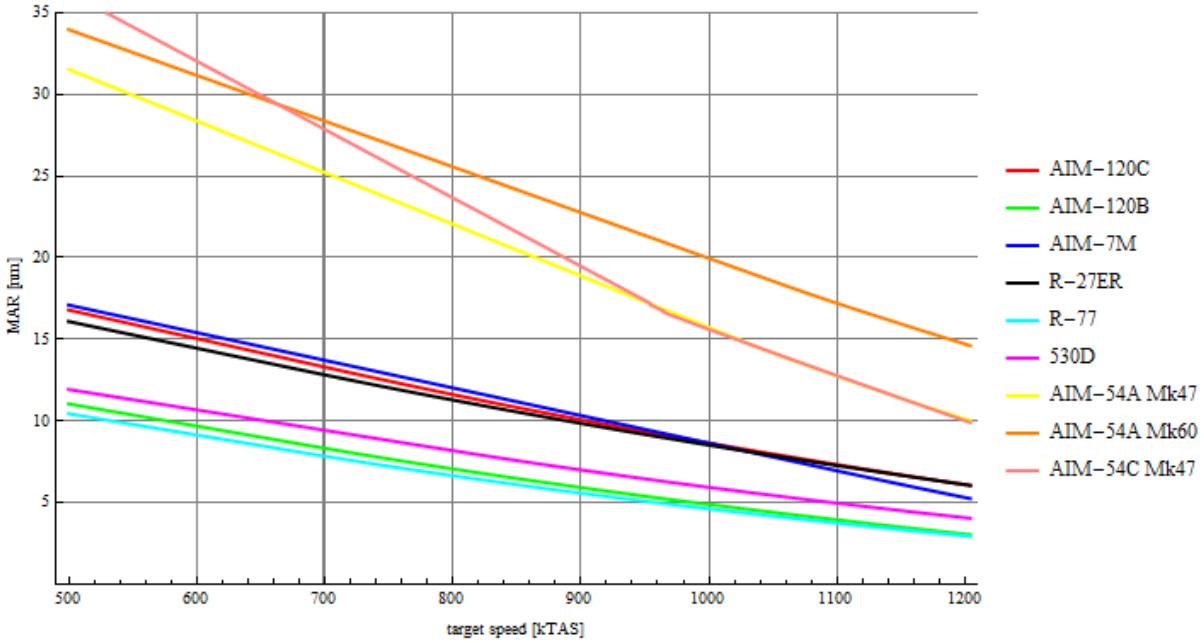


Figure 10: *MAR* from TacView encounters comparison

In figure 10 I've put the *MAR* dependency on the target speed for the TacView experienced scenarios. The three most standoffish missiles in this case are the AIM-120C, AIM-7M and the R-27ER which can reach the 12-nm kill zone for the entire 500-700 kTAS region. Above 1000 kTAS, the AIM-120C and the R-27ER get an advantage over the Sparrow.

The surprise comes from the Super 530D which is the 4th missile based on *MAR*. It starts off at 12 nm for slow targets and at 700 kTAS run-away speed it can still reach 9 to 10 nm. Even as speed increases into the super sonic region, the slope of the *MAR* remains shallow and an 1'100 kTAS target has to become very creative within 5 nm from the Mirage.

The AIM-120B and the R-77 are the least standoffish missiles out there. But as they are highly maneuverable fire-and-forget missiles, the results are not surprising. The fact is that if one finds him/her-self within the *MAR*, chances are he/she cannot beat it by maneuvering alone.

The Phoenix Mk60 is quite good at maintaining maximum *MAR*. However, the Mk47s *MAR* loss is within the field of the other missiles'.

Figure 11 shows the comparison of the degradation of the *MAR* for the whole target speed spectrum. In normal BVR encounters ($\sim$ 700 kTAS target speed), most *MAR*s see a 20% degradation and 25% for the R-77 and the AIM-120B. This really levels the field in terms of initial speed, motor burn and deceleration. The Mk60 Phoenix looses just 15% against a 700-knot target.

The next plot, figure 12 shows the *MAR* to MOR degradation for the target speed spectrum.

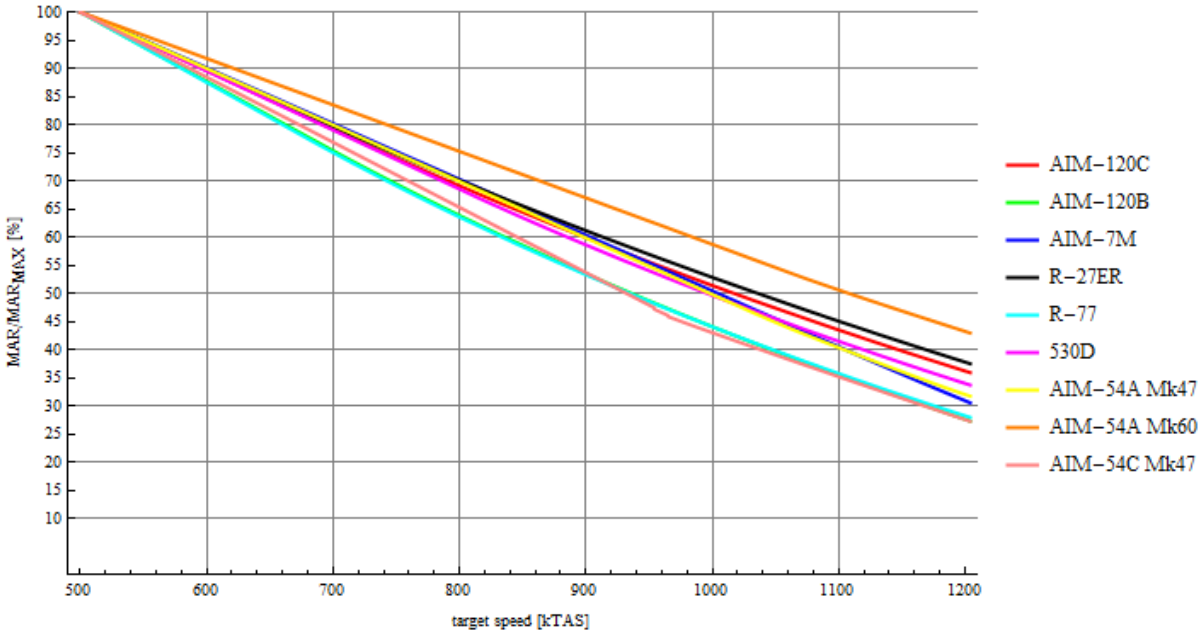


Figure 11: *MAR* from TacView encounters comparison, normalized to the maximum *MAR* comparison

Having the least MOR, the 530D excels at this parameter as for a slow target, even at 25'000 ft and without reaching maximum operational speed, it still kills at over 60% of the MOR. Even pursuing a Mach 2 target, it can still do over 20% of the MOR which is over 4 nm!

The next missile with a nice *MAR*/MOR ratio is the Sparrow. This is the second least MOR value missile but it has a slower degradation than the 530D (leaner slope).

The other 4 missiles have less than 4% *MAR* to MOR degradation throughout the target speed spectrum and achieve the least range compared to their MOR. Nevertheless, these are high MOR missiles and the fact that their degradation slopes are the leanest makes them much more dangerous than the 530 or the Sparrow.

The AIM-54 Phoenix is the worst missile in keeping its' MOR. The advertised ranges are utterly useless as it already starts with an -84% MOR hit and ends up with a less than 5% *MAR* to MOR ratio.

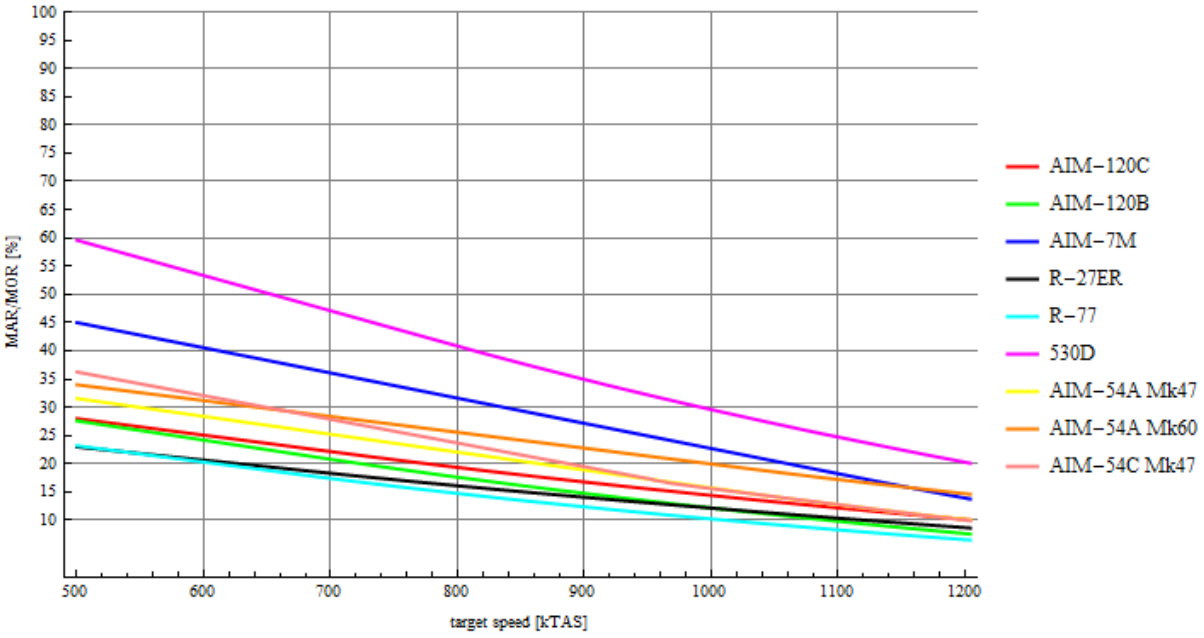


Figure 12: *MAR* from TacView encounters comparison, normalized to the maximum *MAR* comparison

4 Conclusion

The no. 1 BVR scope is **NOT TO MERGE**! As the bandit is getting closer and closer, the *MAR* can be a definite separator between the two combatants. But as I have found out from various encounters, the *MAR* is not a clean cut parameter of a missile. This analysis strove to prove that missiles actually kill the unprepared, as the missile field is pretty much leveled out.

Yes, you can fire the AMRAAM-C and the R27ER at horrendous ranges and put the enemy on the defensive, but that just means he/she can turn away for a few seconds and then come back hot and shoot his/her own as the range has decreased. Leveling the odds in BVR is the no. 2 scope. And if you've also managed to make the bandit loose one of his missiles in the process, the better.

GOOD LUCK!